

IVY MATOBORI

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SKILLS

- **Embedded Systems & Firmware:** Embedded C/C++, FreeRTOS, ESP-IDF, ESP32, bare-metal development, peripheral drivers, calibration procedures, flow and temperature measurement
- **Industrial Automation:** PLC programming, ladder logic, Siemens TIA Portal, HMI configuration, machine control, alarm handling, safety interlocks
- **Interfaces & Connectivity:** I²C, SPI, UART, CAN bus, RS485, Modbus, one-wire, ADC, PWM, GPIO, LoRa, BLE, NFC, WiFi, 4G/LTE
- **Hardware & PCB Design:** Altium Designer, KiCad, Fusion 360, schematic capture, multi-layer PCB layout, MCU & component selection, DFM, PCBA, board bring-up
- **Power & Validation:** Power architecture, regulation, protection, redundancy, low-power design, energy metering, oscilloscope, logic analyzer, system validation, Git, technical documentation

EDUCATION

Jomo Kenyatta University of Agriculture and Technology (JKUAT)

Nairobi, Kenya

Bachelor of Science in Control and Instrumentation Engineering

2019 – 2024

WORK EXPERIENCE

Quepay Limited, Kenya | C++ · FreeRTOS · ESP-IDF · Altium · LTE/BLE/I²C

Feb 2025 – Present

Hardware Designer & Technical Operations Lead

On-site, Nairobi, Kenya

- Sustained a **99%** device uptime SLA across **KES 10M+** in processed transactions, as measured by fleet monitoring, by building the C++ and FreeRTOS firmware integrating M-Pesa, MTN MoMo, and Airtel Money flows over LTE.
- Consolidated **6+** machine verticals onto one controller board, from water and milk ATMs to vending and laundry, by designing the hardware and **4-layer** PCB in Altium Designer, scaling platform volume to **KES 2M+** in **4 months**.
- Specified, procured, installed, and commissioned controller and sensing hardware across **6** field stations, as measured by site acceptance sign-off, validating each installation on site before hand-off to the operator.
- Cut board-level defect rate to **2%** across **300+** units shipped and per-unit validation time to **12 minutes**, by standardising test procedures and leading a **4-person** team across assembly, inspection, and field deployment.
- Drove long-term measurement accuracy on milk ATM dispensers, holding dispensed volume within **±1.8%** of setpoint, by developing **calibration procedures** for flow and temperature instrumentation and troubleshooting instrument failures in the field.
- Held dispense faults below **0.6%** across **250** deployed units, by building fault detection, alarm handling, and safety interlocks into the controller firmware, including over-dispense cutoff and automatic payment reversal on incomplete dispense.
- Reduced field return rate **58%** across **250** installed units, as measured over **12 months** of deployment, by performing board bring-up and signal integrity checks on every revision and root-causing **15** production escalations.

Veno Autobotics, Kenya | Level & Energy Metering · PCB Design · LoRa/BLE/NFC

Apr 2024 – Jan 2025

Embedded Systems Engineer, Embedded Hardware Design

On-site, Nairobi, Kenya

- Shipped **6** IoT products across school, utility, retail, and municipal deployments, by owning embedded hardware design on a **5-engineer** studio team spanning level sensing, energy metering, amusement metering, and public display systems.
- Instrumented liquid level and energy consumption for field deployments, by designing the sensing hardware behind Veno's Level Sensor and VeNode energy meter, scaling metering from a single appliance to a full industrial complex.
- Extended device reach beyond cellular coverage across **4** radio technologies, LoRa, BLE, NFC, and WiFi, by designing low-power RF hardware and selecting the radio per deployment against range, power budget, and site constraints.
- Laid out the smallest PCB in company history at **15 mm** by **20 mm**, by taking client hardware from schematic and layout through prototype assembly to enclosure and packaging design with mechanical engineers.
- Held control loop jitter under **2 ms** on **32 KB** of flash, by developing bare-metal C/C++ firmware implementing **8** peripheral drivers over I²C, SPI, and UART and running end-to-end system verification before design freeze.

Kenya Space Agency, Kenya | EPS · Power Distribution · MCU Selection

Jan 2023 – Mar 2023

Satellite Systems Engineer (EPS)

On-site, Nairobi, Kenya

- Achieved **92%** conversion efficiency against a **5 W** orbit-average power budget, as measured by bench characterisation, by designing the satellite power distribution PCB on a **6-layer** stack-up with **4** regulated rails on redundant paths.
- Cut standby current draw **18 mA** against the baseline design, by leading microcontroller selection for the EPS subsystem and trading **6** candidate parts against power, radiation tolerance, and reliability constraints.
- Carried the EPS design through **2** design reviews within a **3-month** placement, by specifying a redundancy architecture with **6** fault-isolated branches and producing the interface and test documentation.

Gearbox, Kenya | KiCad · Embedded C · Prototyping · Power Management

Jan 2022 – Apr 2022

Embedded Systems Apprentice

On-site, Nairobi, Kenya

- Delivered **3** working embedded demonstrators in **16 weeks**, as measured against **10** functional acceptance criteria per build, by prototyping microcontroller, sensor, and actuator systems and validating each before hand-off.
- Turned **4** boards from concept to tested hardware within a **3-week** cycle, by completing schematic capture, **2-layer** PCB layout, and DFM review in KiCad.
- Brought **6** sensor interfaces online and cut prototype idle power draw **35%**, by developing bare-metal firmware in embedded C and optimising supply rails and sleep modes across **4** boards.